Uniform vanishing-viscosity convergence for a local quadratic mean field game

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Abstract

We prove uniform vanishing-viscosity convergence for the local mean field game system with quadratic Hamiltonian and linear density coupling on a flat torus in dimensions one, two and three. For every smooth nonnegative probability initial density, smooth fixed terminal cost and positive time horizon, the entire family of value functions converges uniformly on each interior time slab to a single continuous function. Initial vacuum is allowed. Averaging finite-energy perturbations of optimal trajectories turns a density-gradient estimate into a modulus of continuity independent of viscosity. A convex variational limit and path calibration then identify the occupied-region value, while homogeneous Hamilton–Jacobi comparison determines its vacuum extension.

1 Introduction

Mean field games with local coupling describe a population whose individual running cost depends on the density at the current position. In the quadratic case, a backward Hamilton–Jacobi equation for the value function is coupled to a forward Fokker–Planck equation for the population density. Positive viscosity represents diffusion in the individual trajectories. For smooth periodic data, classical solutions at each positive viscosity are available [3, Theorem 4.1]. The question here concerns their deterministic limit, rather than their existence at fixed viscosity.

Weak formulations of first-order mean field games with local coupling can be obtained from convex variational problems [2]. Spatial Sobolev estimates for such variational systems are developed in [5]. These methods control the density and value gradient, but uniform convergence of values also requires control of their behavior where the density is small. Tang and Zhang [8, Section 7, problem 2(a)] ask whether the value functions converge locally uniformly as viscosity vanishes. Their local-coupling convergence estimates include density-weighted value bounds, which do not alone control values in vacuum.

We treat the quadratic Hamiltonian with linear local coupling and a fixed terminal cost. Let $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$, with volume one. For smooth data m_0, g and $\nu > 0$, consider

$$\begin{aligned} -\partial_t u_\nu - \nu\Delta u_\nu + \tfrac{1}{2}|Du_\nu|^2 &= m_\nu, \\ \partial_t m_\nu - \nu\Delta m_\nu - \operatorname{div}(m_\nu Du_\nu) &= 0 \qquad \text{in } (0,T)\times\mathbb{T}^d, \end{aligned} \tag{1.1}$$

with

$$m_\nu(0) = m_0, \qquad u_\nu(T) = g. \tag{1.2}$$

The initial density may vanish. Uniform convergence is therefore a statement about values on the whole torus, including regions that density-weighted estimates may not see.

Theorem 1.1. *Let $d \in \{1,2,3\}$, $T > 0$, $m_0 \in C^\infty(\mathbb{T}^d;[0,\infty))$ with $\int_{\mathbb{T}^d} m_0 = 1$, and $g \in C^\infty(\mathbb{T}^d;\mathbb{R})$. Let (u_ν, m_ν) be the classical solutions of (1.1)–(1.2). There is a function $u \in C((0,T)\times\mathbb{T}^d)$ such that, for every $0 < \tau < T/2$,*

$$\|u_\nu - u\|_{L^\infty([\tau,T-\tau]\times\mathbb{T}^d)} \longrightarrow 0 \qquad \text{as } \nu \downarrow 0. \tag{1.3}$$

The limit is the same for the entire family of viscosities.

Thus the uniform-convergence question has an affirmative answer for this quadratic, linear-coupling specialization in dimensions at most three, without a positive lower bound on the initial density or a restriction on the time horizon. The theorem does not address general Hamiltonians, other couplings or higher dimensions. We do not obtain a convergence rate.

The first difficulty is to obtain continuity estimates without a uniform pointwise bound on the density. A differentiated identity bounds its spatial gradient in terms of the value bound. Averaging finite-energy paths that spread from a point closes this estimate. To compare two nearby starting positions, we perturb an optimal trajectory by an independent family of shifts which return to the original trajectory after a short time. The averaged running cost is controlled by the density's space-time norm, and the kinetic increment gives a modulus of continuity independent of viscosity.

The second difficulty is that occupied-region information does not immediately determine a value in vacuum. A heat-semigroup recovery sequence preserves the initial density exactly and identifies a common first-order density and flux. Their finite-action path law calibrates every continuous value limit on the occupied region. On an open region where two limits differ, the common density must be zero; there homogeneous Hamilton–Jacobi comparison proves that the limits coincide. This uses the variational and calibration ideas of [2], with the required specialization proved below.

Section 2 gives the uniform estimates and compactness. Section 3 identifies the density limit and proves calibration. Section 4 completes the value comparison. All spatial integrations use the normalized torus measure; D and Δ act only in space, and dist denotes the shortest-lift torus distance.

2 Uniform estimates

For clarity, the fixed-viscosity classical theory cited above applies in the present normalization. Set $s = \nu t$, $U(s,x) = u_\nu(s/\nu,x)/\nu$ and $n(s,x) = m_\nu(s/\nu,x)$. The equations have unit diffusivity, Hamiltonian $|DU|^2/2$, coupling n/ν^2, horizon νT, terminal value g/ν, and initial density m_0. The coupling is smooth and bounded below on nonnegative densities, so [3, Theorem 4.1 and its regularity conclusion] applies. Smooth data permit the differentiations and endpoint integrations used below. This is only qualitative regularity for each fixed ν.

Throughout the proof $0 < \nu \le 1$, $Q = (0,T)\times\mathbb{T}^d$, and D denotes the spatial gradient. Constants may depend on d, T, m_0, g, and constants denoted C_τ may also depend on τ. They never depend on ν. Mass conservation and positivity give

$$m_\nu \ge 0, \qquad \int_{\mathbb{T}^d} m_\nu(t,x)\,dx = 1.$$

These facts follow by integrating the forward equation and testing its negative part. The resulting inequality $\frac{d}{dt}\|m_\nu^-\|_2^2 \le C_\nu\|m_\nu^-\|_2^2$, with zero initial value, gives the latter assertion.

2.1 Control verification and the density estimate

We first record the control formula used to compare values at nearby points. Let B be a standard d-dimensional Brownian motion. An admissible control is an adapted measurable

process a with $\mathbb{E}\int_t^T |a_s|^2\,ds < \infty$. Positions are projected onto the torus from

$$X_s = x + \int_t^s a_r\,dr + \sqrt{2\nu}(B_s - B_t).$$

Itô's formula, with the periodic lift of u_ν, gives

$$\mathbb{E}\left[g(X_T) + \int_t^T \left(\tfrac{1}{2}|a_s|^2 + m_\nu(s, X_s)\right) ds\right] = u_\nu(t,x) + \tfrac{1}{2}\mathbb{E}\int_t^T |a_s + Du_\nu(s, X_s)|^2 ds. \quad (2.1)$$

The martingale has zero expectation because the test gradient is bounded for fixed ν; all drift terms are integrable by Cauchy–Schwarz. The coefficient $\sqrt{2\nu}$ gives precisely the term $\nu\Delta u_\nu$. See, for example, [6, Theorem 3.7.1] for this version of Itô's formula.

The smooth feedback $a_s = -Du_\nu(s, X_s)$ is admissible and realizes equality. Its process can be constructed by Picard iteration of the integral equation for $X - \sqrt{2\nu}B$, successively on short time intervals. Thus u_ν is the infimum of the left side of (2.1). Applying the same calculation on $[t, s]$, with terminal function $u_\nu(s, \cdot)$, gives

$$u_\nu(t,x) = \inf_a \mathbb{E}\left[u_\nu(s, X_s) + \int_t^s \left(\tfrac{1}{2}|a_r|^2 + m_\nu(r, X_r)\right) dr\right]. \quad (2.2)$$

An independent auxiliary random variable merely averages admissible costs and therefore also supplies an upper bound for these infima.

Set $g_* = \min g$ and $M_\nu = \max_{[0,T]\times\mathbb{T}^d}(u_\nu - g_*)$. Nonnegative running costs imply $u_\nu \geq g_*$. Every optimal trajectory starting at any (t, x) satisfies

$$\mathbb{E}\int_t^T |a_s|^2 ds \leq 2M_\nu. \quad (2.3)$$

The density gradient can be bounded in terms of this value bound. For $u_i = \partial_i u_\nu$, $m_i = \partial_i m_\nu$, differentiating the equations gives

$$u_{i,t} = -\nu\Delta u_i + Du_\nu \cdot Du_i - m_i, \qquad m_{i,t} = \nu\Delta m_i + \operatorname{div}(m_i Du_\nu + m_\nu Du_i).$$

Consequently, periodic integration by parts yields

$$\frac{d}{dt}\int_{\mathbb{T}^d} m_i u_i = -\int_{\mathbb{T}^d} m_i^2 - \int_{\mathbb{T}^d} m_\nu |Du_i|^2.$$

The diffusion terms cancel, as do the two terms containing $m_i Du_\nu \cdot Du_i$. Sum in i and integrate in time:

$$\int_Q (|Dm_\nu|^2 + m_\nu |D^2 u_\nu|^2) = -\int_{\mathbb{T}^d} \Delta m_0\, u_\nu(0) + \int_{\mathbb{T}^d} m_\nu(T)\Delta g \leq \|\Delta m_0\|_1 M_\nu + \|\Delta g\|_\infty. \quad (2.4)$$

Here $\int \Delta m_0 = 0$ removes the constant g_*. No division by the density is involved.

For later use, the mass-one Gagliardo–Nirenberg estimate on the torus is

$$\int_{\mathbb{T}^d} m^{p_d} \leq C_d(1 + \|Dm\|_2^2), \qquad m \geq 0, \quad \int_{\mathbb{T}^d} m = 1, \quad p_d = \frac{2(d+1)}{d}. \quad (2.5)$$

To see the additive term explicitly, multiply the periodic extension by a fixed smooth compact cutoff and apply the Euclidean inequality of [7, Lecture II, (2.2)–(2.3), pp. 125–126]. Writing $Y = \|m\|_{p_d}$, $D_0 = \|Dm\|_2$, $\theta = d/(d+1)$, one obtains $Y \leq C(D_0 + \|m\|_2)^\theta$. Interpolation with mass one gives $\|m\|_2 \leq Y^{(d+1)/(d+2)}$. Absorbing $CY^{d/(d+2)}$ therefore gives $Y \leq C(1 + D_0^\theta)$;

since $\theta p_d = 2$, this proves (2.5). The exceptional Euclidean condition in dimension two requires $\theta < 1$, which holds. The additive constant is necessary even for $m = 1$.

Since $p_d \geq 8/3$ in the dimensions under consideration, (2.4) and (2.5) give, with $p = 8/3$,

$$\|m_\nu\|_{L^p(Q)} \leq C(1 + M_\nu)^{1/p}. \tag{2.6}$$

We close this estimate by averaging paths that spread from their initial point. Choose

$$\beta = \frac{9}{17}, \qquad c_T = \min(1, T^{-\beta}).$$

The conditions needed are $\beta > 1/2$ for finite kinetic energy and $d\beta < p - 1$ for integrability of the running cost. They both hold for $d \leq 3$. If $h = T - t$, take z uniform in the Euclidean unit ball, independent of B, and use

$$a_{t+r} = c_T \beta r^{\beta-1} z, \qquad X_{t+r} = x + c_T r^\beta z + \sqrt{2\nu} B_r, \quad 0 < r < h.$$

Its kinetic cost is at most $C_T h^{2\beta-1}$. For $F \geq 0$, the ball average satisfies

$$\mathbb{E}_z F(Y + \vartheta z) \leq C_d \vartheta^{-d/p} \|F\|_{L^p(\mathbb{T}^d)}, \qquad 0 < \vartheta \leq 1. \tag{2.7}$$

Indeed, its periodic density is bounded by $C_d \vartheta^{-d}$, with support volume at most $C_d \vartheta^d$; the overlap of periodic lifts is bounded. Hölder's inequality gives (2.7). Conditioning on the Brownian position and then applying Hölder in time,

$$\begin{aligned} \mathbb{E} \int_0^h m_\nu(t + r, X_{t+r}) dr &\leq C_T \left(\int_0^h r^{-d\beta/(p-1)} dr \right)^{1/p'} \|m_\nu\|_{L^p(Q)} \\ &\leq C_T h^{a_d} \|m_\nu\|_{L^p(Q)}, \qquad a_d = 1 - \frac{d\beta + 1}{p} > 0. \end{aligned} \tag{2.8}$$

The worst value is $a_3 = 1/34$. The control formula, bounded terminal cost and (2.6) now imply

$$M_\nu \leq A + B(1 + M_\nu)^{3/8}$$

for fixed A, B. Since $3/8 < 1$, this bounds M_ν uniformly. We have proved

$$\|u_\nu\|_\infty + \|m_\nu\|_{L^{8/3}(Q)} + \|Dm_\nu\|_{L^2(Q)} + \|\sqrt{m_\nu} D^2 u_\nu\|_{L^2(Q)} \leq C. \tag{2.9}$$

Integrating the Hamilton–Jacobi equation also gives

$$\tfrac{1}{2} \int_Q |Du_\nu|^2 = T + \int_{\mathbb{T}^d} g - \int_{\mathbb{T}^d} u_\nu(0) \leq C. \tag{2.10}$$

2.2 Spatial and temporal continuity

The averaging argument can be arranged to return to an optimal trajectory after a short time. This permits comparison of nearby initial positions without changing the later cost.

Proposition 2.1. *Let $\alpha = 1/34$. For every $0 < \tau < T$,*

$$|u_\nu(t, x) - u_\nu(s, y)| \leq C_\tau(\operatorname{dist}(x, y)^\alpha + |t - s|^{\alpha/2}), \quad t, s \in [0, T - \tau], \quad x, y \in \mathbb{T}^d. \tag{2.11}$$

Moreover $u_\nu(t, \cdot) \to g$ uniformly as $t \uparrow T$, uniformly in $0 < \nu \leq 1$.

Proof. First fix $t \le T - \tau$. Put $r = \operatorname{dist}(x, y)$ and let q be a shortest lift of $y - x$. For $0 < r < \min(\tau/2, 1/2)$, set $h = r$, and let $\widehat{X}_\ell, \widehat{a}_\ell$ be an optimal process from (t, x). Write $X_s = \widehat{X}_{t+s}$ and $a_s = \widehat{a}_{t+s}$ for elapsed time $0 \le s \le T-t$. Recall $\beta = 9/17$. For an independent unit-ball parameter z, define on $[0, h]$

$$\vartheta_h(s) = h^\beta \left[\frac{s}{h}\left(1 - \frac{s}{h}\right)\right]^\beta, \qquad X_s^z = X_s + (1 - s/h)q + \vartheta_h(s)z.$$

The new trajectory starts at y, uses the same Brownian noise, and equals X_h at time h. Its adapted control is

$$a_s^z = a_s + b_s^z, \qquad b_s^z = -q/h + \vartheta_h'(s)z.$$

After h, use the original process and control. This is an admissible control for each fixed z: the added term is deterministic and square-integrable. In fact,

$$\int_0^h |b_s^z|^2 ds \le C\left(\frac{r^2}{h} + h^{2\beta-1}\right). \tag{2.12}$$

The derivative of $[v(1-v)]^\beta$ has squared endpoint power $2\beta - 2 = -16/17 > -1$, which proves this estimate.

Since X_s is independent of z, (2.7) applies to its shifted position. The time integral now has two endpoints:

$$\int_0^h \vartheta_h(s)^{-d/(p-1)} ds = h^{1-d\beta/(p-1)} \int_0^1 [v(1-v)]^{-d\beta/(p-1)} dv.$$

It is finite because $d\beta/(p-1) \le 81/85 < 1$. With $p = 8/3$ and $a_d = 1 - (d\beta + 1)/p \ge 1/34$, this gives

$$\mathbb{E}\int_0^h m_\nu(t + s, X_s^z) ds \le C h^{a_d} \|m_\nu\|_{L^p(Q)}.$$

All costs after h agree exactly. Subtract the original optimal cost from the averaged competitor cost. Discard the original nonnegative running cost, and bound the kinetic increment $a_s \cdot b_s^z + |b_s^z|^2/2$ by Cauchy–Schwarz, (2.3), (2.9) and (2.12). The result is

$$u_\nu(t, y) - u_\nu(t, x) \le C\left(\frac{r}{\sqrt{h}} + h^{\beta-1/2} + \frac{r^2}{h} + h^{2\beta-1} + h^{a_d}\right).$$

With $h = r$, every exponent is at least $1/34$. Swap x, y, and use the uniform value bound for larger r, to obtain

$$|u_\nu(t, x) - u_\nu(t, y)| \le C_\tau \operatorname{dist}(x, y)^\alpha. \tag{2.13}$$

The cost estimate concerns an averaged family of trajectories, rather than a trace of an L^p function on a single curve.

For time continuity let $0 \le t < s \le T - \tau$, $h = s - t \le 1$. In (2.2), use the spreading control from (2.8), restricted to this time interval. Its kinetic and running costs are bounded by $C(h^{2\beta-1} + h^{a_d})$. The endpoint increment in (2.13) is bounded by

$$C_\tau \mathbb{E} \operatorname{dist}(X_s, x)^\alpha \le C_\tau(h^{\alpha\beta} + h^{\alpha/2}) \le C_\tau h^{\alpha/2}.$$

Here $\nu \le 1$ and the Brownian second moment were used. This bounds $u_\nu(t, x) - u_\nu(s, x)$ from above. For the lower bound discard the nonnegative running cost in (2.2). If $A = \mathbb{E}\int_t^s |a_r|^2 dr$,

$$\mathbb{E} \operatorname{dist}(X_s, x)^\alpha \le C(hA)^{\alpha/2} + C(\nu h)^{\alpha/2}.$$

Minimizing $A/2 - C_\tau h^{\alpha/2} A^{\alpha/2}$ over $A \geq 0$ gives a lower bound $-C_\tau h^{\alpha/(2-\alpha)}$. Since $\alpha/(2-\alpha) > \alpha/2$, the lower time increment has the same modulus. Large time increments follow from the value bound. Together with (2.13), this proves (2.11).

Finally put $h = T - t \leq 1$. The spreading control and the Lipschitz continuity of g give

$$g(x) - C(h + h^{1/2}) \leq u_\nu(t,x) \leq g(x) + C(h^{2\beta-1} + h^{a_d} + h^{1/2}). \tag{2.14}$$

The lower bound follows from the same energy minimization with the Lipschitz exponent one. This proves the terminal assertion. □

Arzelà–Ascoli and a diagonal extraction on $[0, T - 1/n] \times \mathbb{T}^d$ now yield a locally uniform value limit for every vanishing sequence of viscosities. In fact, (2.14) makes the terminal strip uniformly close to g, so the extracted subsequence converges uniformly on the whole closed cylinder. We will use its time-zero trace when identifying the limit.

3 The density limit and calibration

Compactness alone does not determine the value in vacuum. We first identify the common density and flux by a convex minimization problem. A path representation of that pair will then determine every value limit on the occupied region.

3.1 The convex action

For $\lambda \geq 0$, let $\mathcal{K}_\lambda$ consist of narrowly continuous probability curves $\rho : [0,T] \to \mathcal{P}(\mathbb{T}^d)$, with a space-time density $\rho \in L^2(Q)$, and measurable fluxes k of finite kinetic action, satisfying

$$\rho_t - \lambda \Delta \rho + \operatorname{div} k = 0, \qquad \rho(0) = m_0.$$

Narrow continuity means continuity of the integral against each continuous spatial test. The equation includes endpoint traces: for every smooth ψ on the closed cylinder,

$$\langle \psi(T), \rho(T) \rangle - \int_{\mathbb{T}^d} \psi(0) m_0 = \int_Q \rho(\psi_t + \lambda \Delta \psi) + k \cdot D\psi. \tag{3.1}$$

The density is specified at almost every time; the narrow curve supplies the remaining time marginals. Define

$$J_\lambda(\rho, k) = \int_Q \left(\frac{|k|^2}{2\rho} + \frac{\rho^2}{2} \right) + \int_{\mathbb{T}^d} g \, d\rho(T), \qquad I_\lambda = \inf_{\mathcal{K}_\lambda} J_\lambda. \tag{3.2}$$

The kinetic integrand is zero at $(\rho, k) = (0,0)$ and infinite at $\rho = 0, k \neq 0$. In particular, a finite-action flux vanishes in vacuum. The terminal pairing allows a measure, rather than requiring a density at every time.

Lemma 3.1. *Set $j_\nu = -m_\nu Du_\nu$. Then (m_ν, j_ν) minimizes J_ν and*

$$I_\nu = \int_{\mathbb{T}^d} m_0 u_\nu(0) - \frac{1}{2} \int_Q m_\nu^2. \tag{3.3}$$

Its density-square and kinetic integrals are bounded independently of ν, and j_ν is bounded in $L^{4/3}(Q)$.

Proof. Test (3.1) with u_ν, substitute $u_{\nu,t}+\nu\Delta u_\nu = |Du_\nu|^2/2 - m_\nu$, and complete two squares. For every finite-action competitor,

$$J_\nu(\rho,k) = \int_{\mathbb{T}^d} m_0 u_\nu(0) - \frac{1}{2}\int_Q m_\nu^2 + \frac{1}{2}\int_Q (\rho - m_\nu)^2 + \frac{1}{2}\int_Q \frac{|k+\rho Du_\nu|^2}{\rho}. \tag{3.4}$$

At vacuum the last term uses the same perspective convention. The smooth test has bounded derivatives for fixed viscosity, so the weak identity is legitimate for these competitors. Equation (3.4) proves minimization and (3.3).

Let $P_s = e^{s\Delta}$ be the torus heat semigroup, $P_0 = \mathrm{id}$. Its periodic Gaussian kernel is positive, symmetric and of mass one. It commutes with spatial derivatives, preserves mass, and approximates the identity uniformly on continuous functions. These properties follow by summing the Euclidean Gaussian over $\mathbb{Z}^d$; its derivative series converge uniformly on compact positive-time intervals. The competitor $(P_{\nu t}m_0, 0)$ bounds

$$I_\nu \le \frac{T}{2}\|m_0\|_2^2 + \max g.$$

The terminal term in any admissible action is at least $\min g$, so the two nonnegative integral terms are bounded separately. Hölder's inequality gives

$$\int_Q |j_\nu|^{4/3} \le \left(\int_Q \frac{|j_\nu|^2}{m_\nu}\right)^{2/3}\left(\int_Q m_\nu^2\right)^{1/3}, \tag{3.5}$$

which proves the flux bound. □

Proposition 3.2. *There is a unique minimizer (m,j) of J_0 over $\mathcal{K}_0$. As $\nu \downarrow 0$,*

$$I_\nu \longrightarrow I_0, \qquad m_\nu \longrightarrow m \quad \text{strongly in } L^2(Q).$$

Along extracted subsequences the probability curves converge narrowly, uniformly in time, to $m(t)$, and $j_\nu \rightharpoonup j$ in $L^{4/3}(Q)$.

Proof. For every smooth spatial test φ, the continuity equation and Cauchy–Schwarz give

$$|\langle \varphi, m_\nu(t) - m_\nu(s)\rangle| \le \nu|t-s|\|\Delta\varphi\|_\infty + \|D\varphi\|_\infty\sqrt{|t-s|}\left(\int_Q \frac{|j_\nu|^2}{m_\nu}\right)^{1/2}. \tag{3.6}$$

The space of probability measures on the compact torus is compact for narrow convergence. Diagonal extraction against a countable dense family of spatial tests therefore yields a continuous limiting probability curve, with convergence uniform in time. Extract also weak limits in L^2 for the densities and $L^{4/3}$ for the fluxes. One may obtain these by diagonalizing bounded pairings against a countable dense family of dual tests, then using L^p duality [4, Theorem 6.15, p. 190]. Time integration identifies the weak density with the narrow curve at almost every time. Passing to (3.1) proves $(m,j) \in \mathcal{K}_0$ once lower semicontinuity gives finite action. The diffusion term tends to zero since $\nu m_\nu \to 0$ in L^2; the initial marginal remains exactly m_0.

Both nonnegative action terms are weakly lower semicontinuous. For the kinetic term this follows from

$$\frac{|k|^2}{2\rho} = \sup_b \left(k\cdot b - \frac{\rho|b|^2}{2}\right).$$

The integral equals the supremum over smooth bounded fields. To check this without a positivity assumption, take b_N in the direction of k, of magnitude $\min(|k|/\rho, N)$ where $\rho > 0$, and magnitude N where $\rho = 0, k \neq 0$. Its nonnegative dual integrand increases to the perspective, including the infinite vacuum case. Smooth convolution approximates b_N in L^1 and along a subsequence almost everywhere, keeping the bound N. Dominated convergence applies with $N|k| + N^2\rho/2$, and monotone convergence lets $N \to \infty$. Each smooth dual pairing is weakly continuous, proving the assertion. The same argument with $c_N = \min(\rho, N)$ and $\rho^2/2 = \sup_c(\rho c - c^2/2)$ applies to the density square. The convolution approximation is the usual approximate-identity property [4, Theorem 8.14, pp. 242–243]; periodic functions are localized by a compact cutoff, and time extensions are zero. Thus

$$J_0(m, j) \leq \liminf_{\nu \downarrow 0} I_\nu \tag{3.7}$$

along the chosen subsequence, with the terminal term converging narrowly.

For the opposite inequality, take any finite-action $(\rho, k) \in \mathcal{K}_0$ and set

$$\rho_\nu(t) = P_{\nu t}\rho(t), \qquad k_\nu(t) = P_{\nu t}k(t). \tag{3.8}$$

Use $P_{\nu t}\psi(t)$ as the test in the zero-diffusion equation. Since $\partial_t(P_{\nu t}\psi) = P_{\nu t}(\psi_t + \nu\Delta\psi)$, self-adjointness and commutation with D give (3.1) for (ρ_ν, k_ν) with diffusivity ν. At time zero $P_0 = \mathrm{id}$, so $\rho_\nu(0) = m_0$ exactly. For the positive probability kernel, weighted Cauchy–Schwarz gives

$$|P_s k|^2 \leq P_s\rho\, P_s(|k|^2/\rho), \qquad (P_s\rho)^2 \leq P_s(\rho^2).$$

Integration proves that neither integral action term increases. Moreover $\int g\, d\rho_\nu(T) = \int P_{\nu T} g\, d\rho(T) \to \int g\, d\rho(T)$. Consequently

$$\limsup_{\nu \downarrow 0} I_\nu \leq I_0.$$

The set $\mathcal{K}_0$ is nonempty, since $(m_0, 0)$ is admissible. Together with (3.7), this proves attainment and convergence of the minima.

If two minimizers had different densities, the density-square term at their midpoint would decrease the average action by $\frac{1}{8}\int(\rho - \widetilde{\rho})^2$, while the kinetic perspective is convex. Thus the densities coincide. At a common density the kinetic midpoint defect is $\frac{1}{8}\int |k - \widetilde{k}|^2/\rho$, and both fluxes are zero in vacuum. This proves uniqueness. Equality of the space-time densities also identifies their narrow curves at all times by continuity. Finally, convergence of total action and separate lower semicontinuity force convergence of the density-square integrals. Weak convergence and this norm convergence imply strong L^2 convergence. Every vanishing sequence has the same conclusion, so the density convergence and convergence of the minima hold for the entire family. □

Let u be any uniform accumulation point on the closed cylinder supplied by Proposition 2.1. Passing in (3.3) gives

$$\int_{\mathbb{T}^d} m_0 u(0) = I_0 + \frac{1}{2}\int_Q m^2. \tag{3.9}$$

The gradient estimate (2.10) identifies $Du \in L^2(Q)$ as the weak limit of a further subsequence of gradients. Convex lower semicontinuity against nonnegative tests in the Hamilton–Jacobi equation then gives

$$-u_t + \tfrac{1}{2}|Du|^2 \leq m \quad \text{in distributions on } Q. \tag{3.10}$$

The term $\nu\Delta u_\nu$ tends to zero distributionally, and the terminal trace is $u(T) = g$. Both (3.9) and (3.10) hold for every value accumulation point with the same pair (m, j).

3.2 A common family of calibrated paths

We use a path representation of the finite-action continuity equation. Its construction and the subsequent calibration are the quadratic specializations of the path arguments in [2, Section 4.3]. We give the details needed here, without invoking that paper's general MFG existence or uniqueness theorem.

Lemma 3.3. *There is a probability law η on $\Gamma = C([0,T];\mathbb{T}^d)$, with the uniform metric, such that*

$$(e_t)_{\#}\eta = m(t) \quad (0 \le t \le T), \qquad \int_\Gamma \int_0^T |\dot\gamma|^2 \, ds \, d\eta(\gamma) \le \int_Q \frac{|j|^2}{m}. \tag{3.11}$$

Here $e_t(\gamma) = \gamma(t)$. The law is concentrated on absolutely continuous paths with square-integrable velocities, using local Euclidean lifts on the torus.

Proof. For $\delta > 0$, smooth only in space with the torus heat semigroup $P_\delta = e^{\delta\Delta}$: $m^\delta = P_\delta m$, $j^\delta = P_\delta j$. The positive heat kernel and mass one give $m^\delta(t,x) \ge \kappa_\delta > 0$. All spatial derivatives of m^δ are uniformly bounded in time, and those of $j^\delta(t)$ are bounded by $C_\delta \|j(t)\|_1$. Thus $b^\delta = j^\delta/m^\delta$ has integrable-in-time supremum and spatial Lipschitz bounds. Its integral equation has a unique flow, by Picard iteration on finitely many intervals where the integrated Lipschitz bound is below $1/2$. Reversing time supplies the inverse flow.

The flow transports $m^\delta(0)$ to $m^\delta(t)$. Indeed, $\partial_t m^\delta = -\operatorname{div} j^\delta$ has an integrable-in-time smooth-spatial right side. A smooth terminal test composed with the backward flow satisfies $\zeta_t + b^\delta \cdot D\zeta = 0$; differentiating $\int \zeta m^\delta$ gives zero. The same identity along the flow identifies every terminal test integral of the push-forward. The required time derivatives are absolutely continuous, so the product rule follows by integration of their L^1 derivatives.

Let η^δ be the law of these flow paths, with initial distribution $m^\delta(0)$. Weighted Cauchy–Schwarz for the heat kernel yields

$$\int_\Gamma \int_0^T |\dot\gamma|^2 ds \, d\eta^\delta = \int_Q \frac{|j^\delta|^2}{m^\delta} \le \int_Q \frac{|j|^2}{m}.$$

The path energy, infinite outside the finite-energy paths, is lower semicontinuous and has compact sublevels. One convenient expression is

$$\mathcal{E}(\gamma) = \sup_{0=t_0<\cdots<t_n=T} \sum_i \frac{\operatorname{dist}(\gamma(t_i),\gamma(t_{i-1}))^2}{t_i - t_{i-1}}.$$

For an absolutely continuous path, Cauchy–Schwarz bounds this expression by its energy. Conversely, polygonal interpolants on vanishing-mesh partitions have bounded L^2 velocity whenever the displayed supremum is finite. On local torus charts, weak extraction of these velocities and uniform convergence of the interpolants give an absolutely continuous limit lift. Interval averages of its velocity converge in L^2, so the supremum equals its energy. This also proves lower semicontinuity, since each partition expression is continuous. Its sublevels have modulus $\operatorname{dist}(\gamma(t),\gamma(s)) \le \sqrt{R|t-s|}$ at energy R, and are compact by Arzelà–Ascoli.

Markov's inequality therefore makes the laws tight. The path space is complete and separable, so Prokhorov compactness and the lower-semicontinuity inequality for integrals give a narrow limit η with the asserted energy bound; see [1, Appendix C, Theorems C.9 and C.12, p. 430 of the author manuscript]. The evaluation maps are continuous and $P_\delta m(t) \to m(t)$ narrowly for every time. They give all the marginals in (3.11). □

Proposition 3.4. *Every value accumulation point u satisfies, on one η-full set,*

$$u(t,\gamma(t)) = g(\gamma(T)) + \int_t^T \left(\tfrac{1}{2}|\dot\gamma(s)|^2 + m(s,\gamma(s))\right) ds, \qquad 0 \le t \le T. \tag{3.12}$$

In particular, any two value accumulation points u, v agree $m(t)$-almost everywhere at every time, and agree almost everywhere on the occupied region $\{m > 0\} \subset Q$.

Proof. Convolve (3.10) with a positive smooth unit-mass space-time kernel. Denote the resulting functions by u_ϵ, m_ϵ; these differ from the spatial heat regularizations used in Lemma 3.3. Convexity gives

$$-u_{\epsilon,t} + \tfrac{1}{2}|Du_\epsilon|^2 \le m_\epsilon$$

on each compact interior time interval for small ϵ. Along an absolutely continuous finite-energy path the chain rule and completion of a square give

$$-\frac{d}{ds}u_\epsilon(s,\gamma(s)) \le m_\epsilon - \tfrac{1}{2}|Du_\epsilon|^2 - Du_\epsilon \cdot \dot\gamma \le m_\epsilon + \tfrac{1}{2}|\dot\gamma|^2.$$

The endpoints pass by local uniform convergence of u_ϵ. The running cost passes for almost every path because

$$\begin{aligned}\int_\Gamma \int_0^T |m_\epsilon(s,\gamma(s)) - m(s,\gamma(s))|\, ds\, d\eta &= \int_Q m|m_\epsilon - m| \\ &\le \|m\|_2 \|m_\epsilon - m\|_2 \longrightarrow 0. \end{aligned} \tag{3.13}$$

Here the time extension of m is zero; periodic spatial convolution is justified by localization of the extension. Restrict the smooth inequality to interior intervals before taking the limit. A subsequence gives convergence of the absolute path integrals almost surely. Pass first for rational endpoints, then for all endpoints by continuity and integrability of the path cost, finally including zero and T by the value traces. This gives, on one full set,

$$u(a,\gamma(a)) \le u(b,\gamma(b)) + \int_a^b \left(\tfrac{1}{2}|\dot\gamma|^2 + m(s,\gamma(s))\right) ds \quad (0 \le a < b \le T). \tag{3.14}$$

The cost is integrable: its expectation is at most $\frac{1}{2}\int |j|^2/m + \int m^2$. A Borel representative of m may be chosen arbitrarily; the occupation identity in (3.13) makes changes on space-time null sets irrelevant to almost every path-time.

For $a = 0, b = T$, the expected right side of (3.14) is at most

$$\frac{1}{2}\int_Q \frac{|j|^2}{m} + \int_Q m^2 + \int g\, dm(T) = I_0 + \frac{1}{2}\int_Q m^2 = \int_{\mathbb{T}^d} m_0 u(0),$$

by (3.9). The inequality supplies the reverse bound, so its nonnegative pathwise gap has zero mean and vanishes almost surely. Split that zero gap at t. Both partial gaps are nonnegative and therefore vanish, proving (3.12) for every t on the same full set.

Every other accumulation point satisfies the identical formula for the same m, j, η. Intersect the two full sets and push forward by e_t to get equality $m(t)$-almost everywhere. For almost every time this marginal has density $m(t,x)$; Fubini then gives $\int_Q m|u - v| = 0$, proving the last assertion. □

4 Uniqueness of the value limit

The equality on the occupied region leaves only vacuum to consider. Every accumulation point is a viscosity supersolution of the homogeneous equation

$$-u_t + \tfrac{1}{2}|Du|^2 \ge 0. \tag{4.1}$$

For completeness, if a smooth test ϕ touches u from below at an interior point, make the touch strict and take nearby interior minimizers of $u_\nu - \phi$. There $u_{\nu,t} = \phi_t$, $Du_\nu = D\phi$ and $\Delta u_\nu \geq \Delta\phi$. Hence

$$-\phi_t - \nu\Delta\phi + \tfrac{1}{2}|D\phi|^2 \geq m_\nu \geq 0.$$

Uniform convergence permits passage to the touching point, proving (4.1). A fourth-power local perturbation removes the strict-touch restriction. No pointwise limiting density is used.

Let u, v be two accumulation points and set $O = \{(t,x) \in Q : u(t,x) > v(t,x)\}$. Proposition 3.4 gives $m = 0$ almost everywhere in O. Equation (3.10) therefore makes u a distributional subsolution of the homogeneous Hamilton–Jacobi equation there. It is also a viscosity subsolution. Indeed, on compact subsets of O, positive convolution and convexity give smooth subsolutions converging uniformly to u. Interior maxima against a smooth upper test pass the subsolution inequality to the limit. The gradient $Du \in L^2$ justifies this convolution.

We show directly that O is empty. This is the homogeneous comparison used for the vacuum extension in [2, proof of Theorem 4.2]. Suppose $u(t_0, x_0) > v(t_0, x_0)$. Choose $0 < a < t_0$ and small $\varepsilon > 0$ so that

$$w(t,x) = u(t,x) - \frac{\varepsilon}{t-a}$$

still exceeds v at (t_0, x_0). On $O \cap \{t > a\}$ it is a strict viscosity subsolution:

$$-w_t + \tfrac{1}{2}|Dw|^2 \leq -\frac{\varepsilon}{(t-a)^2}.$$

The positive maximum of $w - v$ stays away from a, where w tends below v, from T, where $u = v = g$, and from the interior boundary of O, where $u = v$. On a compact time interval containing this maximum, maximize

$$w(t,x) - v(s,y) - \frac{\operatorname{dist}(x,y)^2 + |t-s|^2}{2\delta}.$$

Boundedness and uniform continuity force the separation to zero as $\delta \downarrow 0$, while the limiting difference remains positive. The time coordinates are therefore interior, and the w-test point is in O. For small δ, the shortest lift is unique and the penalty is smooth there. The upper test for w and lower test for v have the same derivatives

$$q = (t-s)/\delta, \qquad p = (x-y)/\delta$$

in these lifts. They would give simultaneously

$$-q + \tfrac{1}{2}|p|^2 \leq -\frac{\varepsilon}{(t-a)^2} < 0, \qquad -q + \tfrac{1}{2}|p|^2 \geq 0,$$

a contradiction. The gradients need not be bounded: the Hamiltonian is independent of time and position and cancels exactly. No initial value condition for the first-order value function is required. Thus $u \leq v$; interchanging them proves equality.

Proof of Theorem 1.1. Every vanishing sequence has a uniformly convergent subsequence on compact interior slabs, and its limit is continuous by Proposition 2.1. The argument above identifies every accumulation point with the same function u. If convergence failed on a fixed slab, a sequence with uniform error bounded below by a positive number would admit a subsequence converging to that same u, a contradiction. The single function obtained this way works for every $0 < \tau < T/2$, which proves the entire-family assertion. □

AI use

AI tools were used in developing the argument, drafting and revising the exposition, checking references and validating the document. Two additional AI contexts reviewed a frozen draft for reader comprehension and mathematical objections. The manuscript is submitted for independent mathematical review.

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